

Course Syllabus

010153523 Algorithms and Data Structures

Semester 1 / Academic Year 2026

Department of Electrical and Computer Engineering (ECE)

Faculty of Engineering, King Mongkut's University of Technology North Bangkok

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Course Information

Course Number:	010153523
Credits:	3 (3–0–6)
Lecture Hours:	3 hours per week (including class exercises)
Course Type:	Required
Prerequisites:	010153002 Computer Programming
Textbook:	Goodrich, M. T., Tamassia, R., & Goldwasser, M. H. <i>Data Structures and Algorithms in Java</i> , 6th Edition. John Wiley & Sons, 2014.

Course Description

This course introduces students to the fundamental data structures and algorithms used in computer programming. Topics include linear structures such as *lists*, *stacks*, and *queues*, and non-linear structures such as *trees* and *graphs*. The course also covers core algorithmic techniques for *sorting*, *searching*, and *recursion*, and emphasises the selection of appropriate data structures and algorithms for problem solving.

Course Learning Outcomes (CLOs)

Upon successful completion of this course, students will be able to:

1. Demonstrate a good understanding of linear data structures, including arrays, linked lists, stacks, and queues.
2. Implement abstract data types effectively.
3. Understand non-linear data structures, including trees and graphs.
4. Select and use appropriate data structures to solve specific problems.
5. Understand and implement sorting, searching, and hashing algorithms.
6. Choose and apply appropriate algorithms to solve specific problems.

Course Topics

- Arrays
- Stacks, Queues, and Deques
- Lists and Iterators
- Trees and Binary Trees
- Priority Queues and Heaps
- Maps and Hash Tables
- Search Trees

- Sorting and Searching
- Graphs

Weekly Schedule

Week	Topic	Reading
1	Course Introduction & Algorithm Analysis	Ch. 1, 4
2	Arrays	Ch. 3
3	Singly Linked Lists	Ch. 3
4	Doubly Linked Lists & Stacks	Ch. 3, 6
5	Queues, Deques & Lists with Iterators	Ch. 6, 7
6	Recursion	Ch. 5
—	Midterm Examination	—
7	Trees & Binary Trees	Ch. 8
8	Priority Queues & Heaps	Ch. 9
9	Maps & Hash Tables	Ch. 10
10	Binary Search Trees	Ch. 11
11	Sorting: Mergesort & Quicksort	Ch. 12
12	Graphs & Shortest Paths	Ch. 14
—	Final Examination	—

Assignment: Bangkok Transit (Graph project) — released after Week 12, due during the final examination period.

Assessment and Grading

Assessment Tools

- Examinations — 80%
- Assignments — 20%

Grading Scale

81–100	A	51–60	C
76–80	B+	46–50	D+
71–75	B	40–45	D
61–70	C+	0–39	F

Course Materials

- Lecture slides: ruslylove.github.io/datastruct
- PDF export of slides: [slides.pdf](#)
- Source repository: github.com/ruslylove/datastruct

This syllabus is subject to minor adjustments at the instructor's discretion to accommodate the academic calendar and class progress.